

DOMINO'S PIZZA SALES ANALYTICS

Power BI Project Documentation

Business Intelligence • DAX • Data Modeling • Power Query ETL • Sales & Peak-Hour Analytics

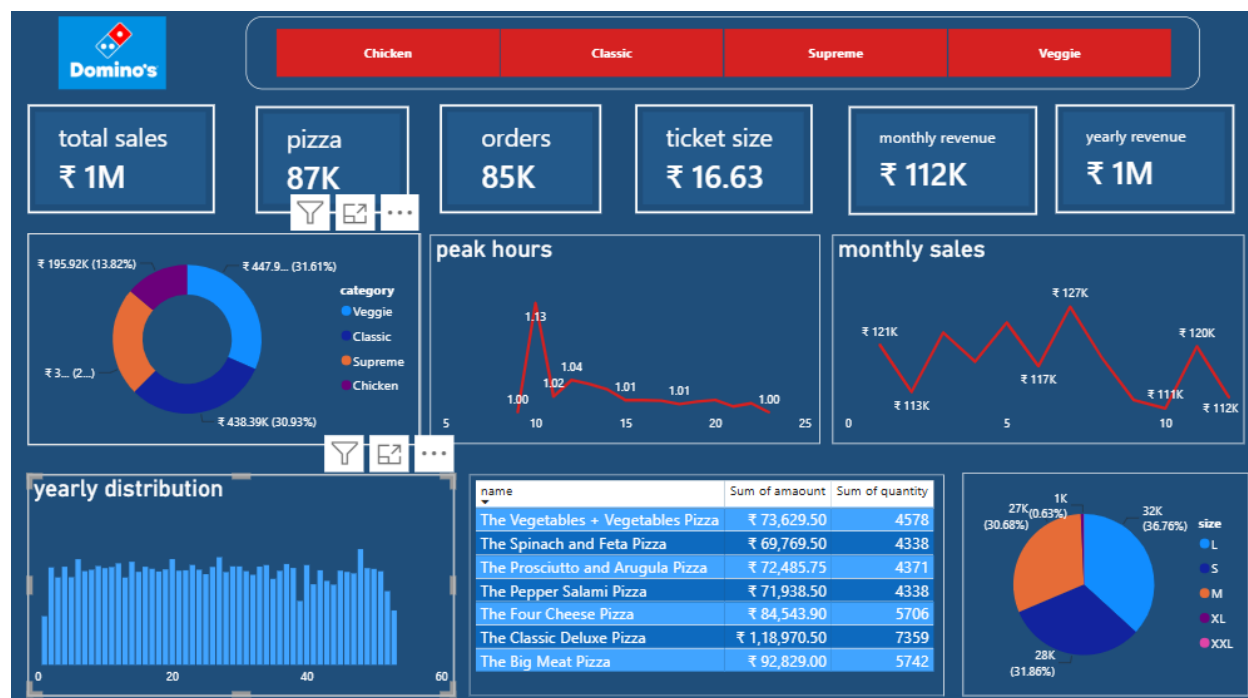


Figure 1 — Power BI Dashboard: Domino's Pizza Sales Overview

1. Project Overview

1.1 Introduction

This project is a Power BI-based sales analytics solution developed for a Domino's Pizza business. The report brings together order, order-detail, pizza, pizza-type, and calendar data to provide a consolidated view of sales performance, product mix, and customer ordering behaviour.

The project focuses on transforming raw transactional pizza-order data into useful analytical information through Power Query ETL, data modeling, calculated fields, and interactive dashboard reporting. The analysis covers total sales, order volume, ticket size, category and size-wise performance, peak ordering hours, monthly revenue trends, and yearly order distribution.

1.2 Project Objectives

- Analyze overall Domino's Pizza sales performance using Power BI.
- Measure total sales, total pizzas sold, total orders, and average ticket size.
- Track monthly and yearly revenue trends.
- Analyze sales by pizza category (Chicken, Classic, Supreme, Veggie).
- Analyze sales by pizza size (S, M, L, XL, XXL).
- Identify peak ordering hours to support staffing and operations planning.
- Rank top-performing pizzas by amount and quantity sold.
- Build a relational data model connecting orders, order details, pizzas, pizza types, and a calendar table.
- Use Power Query Editor to clean, merge, and transform raw CSV order data.

- Create an interactive, filterable dashboard with category-based slicers.

1.3 Scope of the Project

The scope of this project covers descriptive and diagnostic business analytics for a pizza retail business. It focuses on understanding sales patterns, product and size performance, and order-timing behaviour. The project is not presented as a predictive machine-learning system.

2. Dataset Information

2.1 Dataset / Table Description

Table	Purpose in Analysis
orders	Order-level date and hour information for each transaction.
order_details	Line-item level data — order_id, pizza_id, and quantity ordered.
pizzas	Pizza-level information — pizza_id, pizza_type_id, size, and price.
pizza_types	Pizza master data — category, ingredients, and pizza name.
calender	Dedicated date dimension for month, month number, quarter, and week-of-year analysis.

2.2 Data Model / Schema

The report uses a star-schema style model. The orders and order_details tables act as fact tables, while pizzas, pizza_types, and calender act as dimension/lookup tables. Orders connect to order_details, pizzas connect to both order_details and pizza_types, and calender connects to orders for time-based filtering.

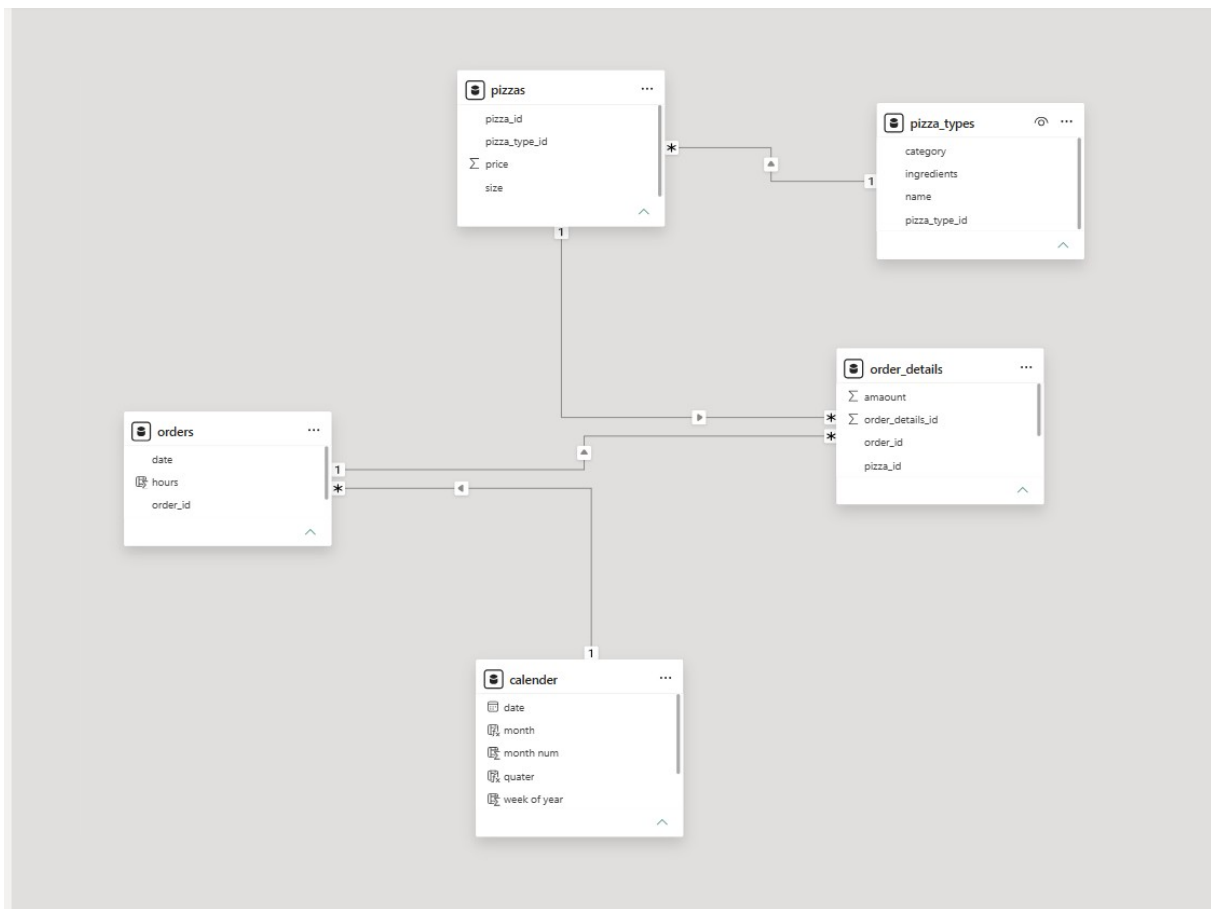


Figure 2 — Power BI Model View: Table Relationships (pizzas, pizza_types, order_details, orders, calender)

2.3 Tables Used in the Report

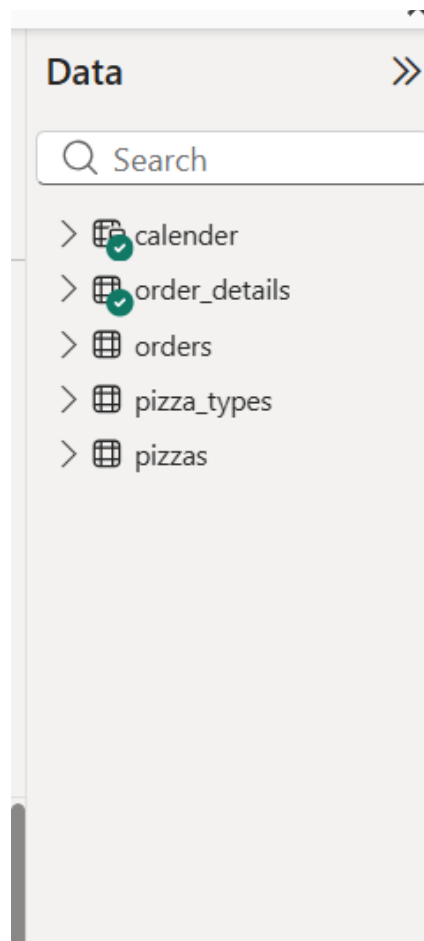


Figure 3 — Data Pane: calender, order_details, orders, pizza_types, pizzas

2.4 Initial Data Assessment

- The Power BI model was reviewed at table and field level before dashboard development.
- Key fields — quantity, price, size, and category — were reviewed for correct data type.
- Relationships between fact and dimension tables were validated in Model View.
- A calendar table was built to support consistent month/quarter/year based reporting.

3. Data Preparation (Power Query ETL)

Raw order data was imported as CSV and processed entirely in Power Query Editor before loading into the Power BI data model. The applied steps transform raw transactional rows into an analysis-ready order_details table.

3.1 Applied ETL Steps — order_details Query

Step	Purpose
Source / Navigation	Connected to the raw CSV order-details source file.
Imported CSV / Promoted Headers	Loaded the file and promoted the first row to column headers.
Changed Type	Set correct data types for order_details_id, order_id, pizza_id, and quantity.
Merged Queries	Merged order_details with the pizzas table to bring in size and price.
Expanded pizzas	Expanded the merged pizzas columns (pizzas.size, pizzas.price) into order_details.

Added Custom	Added a custom "amount" column (quantity × price) to calculate line-item revenue.
Changed Type1	Formatted the new amount column as Currency for reporting.

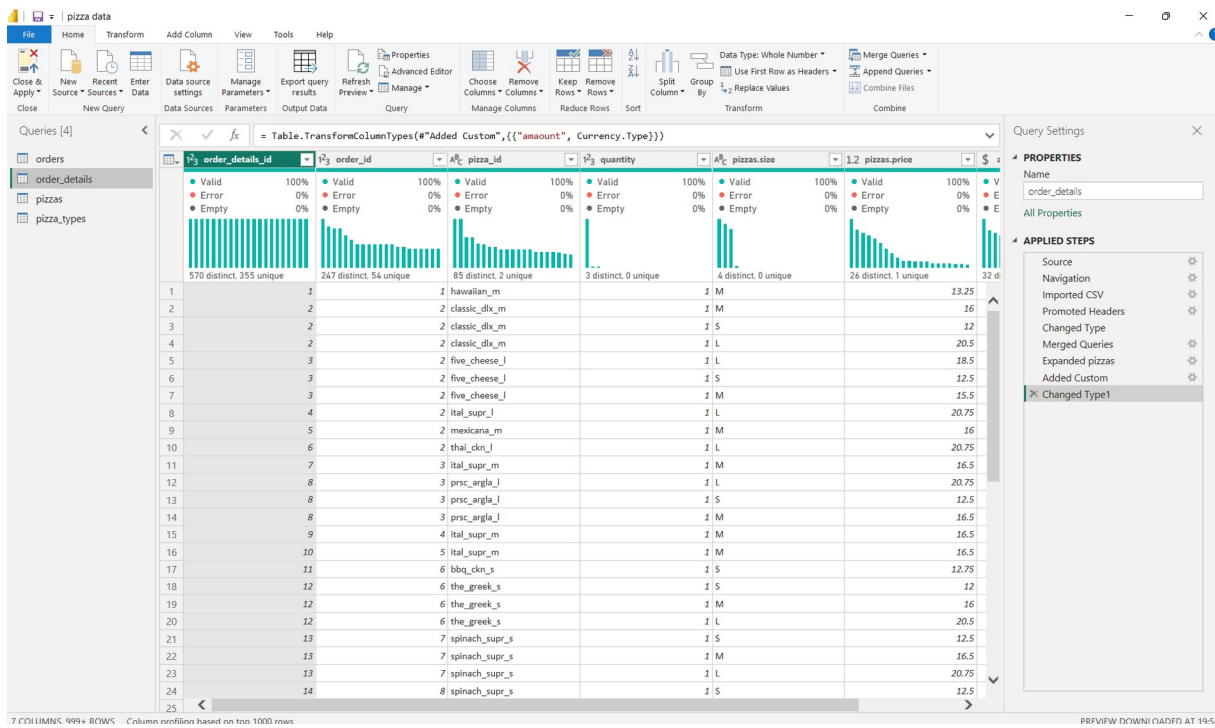


Figure 4 — Power Query Editor: order_details Table with Merged pizzas Data and Applied Steps

3.2 Fields Created / Used

- amount — calculated custom column (quantity × price) used for all revenue-based measures.
- pizzas.size and pizzas.price — brought into order_details via merge & expand from the pizzas table.
- Calendar attributes (month, month number, quarter, week of year) — used for time-based trend analysis.

4. Business Analysis & Insights

4.1 Sales Performance

- Overall sales performance and total revenue tracking (Total Sales ≈ ₹1M).
- Order volume and transaction activity (Orders ≈ 85K, Pizzas Sold ≈ 87K).
- Average ticket size per order (≈ ₹16.63).
- Monthly and yearly revenue tracking.

4.2 Category Performance

- Sales share by pizza category — Classic, Supreme, Veggie, and Chicken — visualised in a donut chart.
- Category-based slicer buttons allow filtering the full dashboard by pizza category.

4.3 Size-Wise Performance

- Order distribution across pizza sizes (S, M, L, XL, XXL) visualised in a pie chart.
- Supports identifying the most popular size for inventory and pricing decisions.

4.4 Peak Hours Analysis

- Hourly order pattern (5:00–25:00 scale) highlights the busiest ordering hour, peaking sharply around hour 13.
- Supports staffing and kitchen-capacity planning during high-demand hours.

4.5 Top-Performing Pizzas

- Ranked table of pizzas by Sum of Amount and Sum of Quantity.
- The Classic Deluxe Pizza and The Big Meat Pizza are among the top revenue generators.

4.6 Time-Based Analysis

- Monthly sales trend line highlights fluctuation across the year (₹111K–₹127K range).
- Yearly distribution chart shows order volume spread across the reporting period.

5. Power BI Dashboard & Report Structure

The final Power BI report combines KPI cards, category slicers, and multiple analytical visuals on a single interactive page, styled with the Domino's brand colour theme.

5.1 KPI Cards



Figure 5 — KPI Cards: Total Sales, Pizzas Sold, Orders, Ticket Size, Monthly Revenue, Yearly Revenue

5.2 Top Pizzas Table

name	Sum of amaount	Sum of quantity
The Vegetables + Vegetables Pizza	₹ 73,629.50	4578
The Spinach and Feta Pizza	₹ 69,769.50	4338
The Prosciutto and Arugula Pizza	₹ 72,485.75	4371
The Pepper Salami Pizza	₹ 71,938.50	4338
The Four Cheese Pizza	₹ 84,543.90	5706
The Classic Deluxe Pizza	₹ 1,18,970.50	7359
The Big Meat Pizza	₹ 92,829.00	5742

Figure 6 — Top Pizzas by Amount and Quantity Sold

5.3 Report View in Power BI Desktop

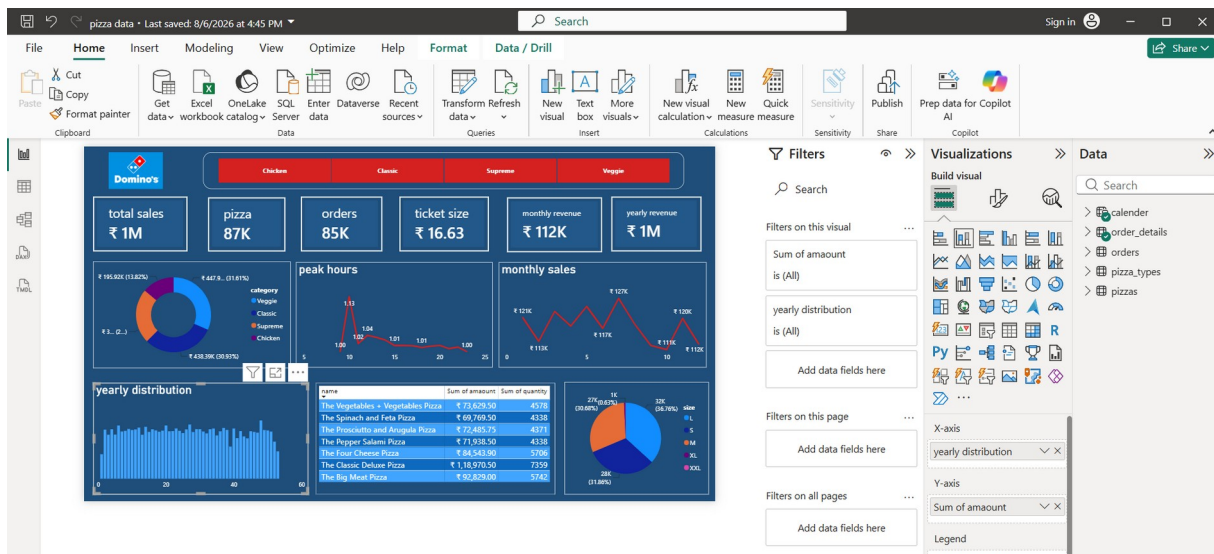


Figure 7 — Power BI Desktop: Full Report Canvas with Filters, Visualizations, and Data Pane

6. Key Performance Indicators (KPIs)

KPI	Purpose
Total Sales	Measures overall pizza sales revenue ($\approx ₹1\text{M}$).
Total Pizzas Sold	Measures total pizza units sold ($\approx 87\text{K}$).
Total Orders	Measures total order volume ($\approx 85\text{K}$).
Ticket Size	Measures average revenue per order ($\approx ₹16.63$).
Monthly Revenue	Measures current-month revenue ($\approx ₹112\text{K}$).
Yearly Revenue	Measures full-year revenue ($\approx ₹1\text{M}$).
Sales by Category	Measures revenue share across Chicken, Classic, Supreme, and Veggie.
Sales by Size	Measures order share across S, M, L, XL, and XXL sizes.
Peak Hours	Identifies the busiest hour(s) of the day for order volume.

7. Conclusion

This Domino's Pizza Sales Analytics project demonstrates the use of Microsoft Power BI, Power Query, and data modeling to transform raw transactional order data into an interactive business intelligence solution. The project combines orders, order details, pizzas, pizza types, and a custom calendar table to provide a consolidated view of sales, product-category performance, size-wise demand, and peak ordering hours.

The ETL process — merging and expanding the pizzas table into order_details, adding a calculated amount column, and building a relational star-schema model — strengthens the analytical foundation of the report. The resulting dashboard helps stakeholders understand sales trends, top-performing products, and operational demand patterns from a single, interactive Power BI report.